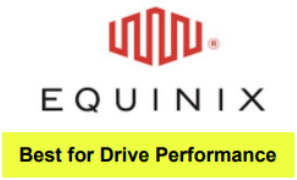
Getting the Bare Metal

After succeeded on deploying Binance private node on AWS EC2 in 2021, I made my way on learning new things by deploying Solana private node in early 2022. This time I do it in non-virtualized pure Bare-Metal environment (it was a challenge for myself). Applied and approved to became part of Solana Server Program to contribute to Solana Network, I made a deal with Equinix to get their Bare-Metal prepared for me.

Ref: *https://solana.org/server-program*`



CONFIG_EQ1

Equinix - w3amd.75xx24c.256.8160.x86

- **CPU** 24 cores @3.0 GHz (AMD EPYC)
- **Storage** 2 x 240 GB SDD + 2 x 3.8TB NVMe
- **Memory** 256GB
- **Network** 2x 25 Gbps w/ 64TB of bundled outbound transfer (Inbound FoC) OS: Ubuntu

The plan was to made a simple business model with Solana Node, Discord community, and Solana NFT. I made Discord Community of Solana NFT Minters and finding its members who needs dedicated and private RPC connection to my private Solana Node. I will later give them permission by whitelisting their IP into my node firewall and NGINX if they already bought my deployed Solana NFT (All of them conducted automatically).

Ref: *https://docs.solana.com/running-validator/validator-reqs*

With the specification above, I put all the services into single Bare Metal machine, including 4 webservices, solana node service, and around 10 RDP VMs, all routed with NGINX reverse proxy for public access.

Installing the Node

First service which needs to be deployed was the Solana validator service itself, as we can see on the server specification, by default it has 2x SSD + 2xNVMe. In Solana requirements, it needs 2 separated high performance disk for the Accounts and the Ledger, so I did allocate 2 partition each from both NVMes. To do this, I need to reformat both of the NVMes to GPT partition from MBR (default), because in MBR partition only has maximum size of 2TB.

Below is how I launch and configure my solana-validator upon launching, took multiple trial & errors to find the optimal configuration for the hardware I used.

***Note**: The configuration may vary and depends on hardware used as Validator/Node*

***solana-validator --ledger** /mnt/ledger*

***--identity** ./validator-keypair.json*

***--trusted-validator** 7Np41oeYqPefeNQEHSv1UDhYrehxin3NStELsSKCT4K2*

***--trusted-validator** GdnSyH3YtwcxFvQrVVJMm1JhTS4QVX7MFsX56uJLUfiZ*

***--trusted-validator** DE1bawNcRJB9rVm3buyMVfr8mBEoyyu73NBovf2oXJsJ*

***--trusted-validator** CakcnaRDHka2gXyfbEd2d3xsvkJkqsLw2akB3zsN1D2S*

***--expected-genesis-hash** 5eykt4UsFv8P8NJdTREpY1vzqKqZKvdpKuc147dw2N9d*

***--entrypoint** entrypoint.mainnet-beta.solana.com:8001*

***--entrypoint** entrypoint2.mainnet-beta.solana.com:8001*

***--entrypoint** entrypoint3.mainnet-beta.solana.com:8001*

***--entrypoint** entrypoint4.mainnet-beta.solana.com:8001*

***--entrypoint** entrypoint5.mainnet-beta.solana.com:8001`*

--no-voting

***--snapshot-interval-slots** 500*

***--maximum-local-snapshot-age** 500*

***--rpc-port** 7000*

***--dynamic-port-range** 8000-8020*

***--no-port-check --wal-recovery-mode** skip_any_corrupted_record **--enable-rpc-transaction-history --enable-cpi-and-log-storage --snapshot-compression none --require-tower --full-rpc-api --no-wait-for-vote-to-start-leader --no-poh-speed-test --accounts** /mnt/account **--limit-ledger-size** 50000000 **--account-index-limit-mb** 50000*

Config Breakdown

Solana Validator receive multiple arguments to launch, I will break down the most important in this project.

--ledger /mnt/ledger : I mounted one of the NVMe to /mnt/ledger as the ledger patition

--no-voting : I used this argument because the node was made for RPC connection not for staking

--identity ./validator-keypair.json : In order to create a node, Solana requires to make an account or a wallet for it, even we will not stake on the node.

--rpc-port 7000 : I set the RPC port to 7000, any RPC request made need to be delivered on port 7000

--dynamic-port range 8000-8020 : Solana node needs to open these ports in order to share and link with other node in the world, so I made sure the ufw (firewall) doesn't block those.

--enable-rpc-transaction-history : This argument will make the node able to get old transaction data, in case the user requesting info for transaction data by the tx hash.

--full-rpc-api: Enable all API functionality of solana-validator.

--accounts /mnt/account : I mounted one of the NVMe to /mnt/account as the account patition

--account-index-limit: This was one of the crucial one, in order to process the user request faster, indexing the account data into RAM was needed, otherwise the node will have extra slow latency upon user request.

```
Your node is behind by 0 slots
Your node is behind by 0 slots
Your node is behind by 0 slots
Your node is behind by 0 slots
Your node is behind by 0 slots
```

This script utilizing **GET /health** request outputting result on the left indicates that the node is running and ready to receive request.

Ref: *https://docs.solana.com/running-validator/validator-start*

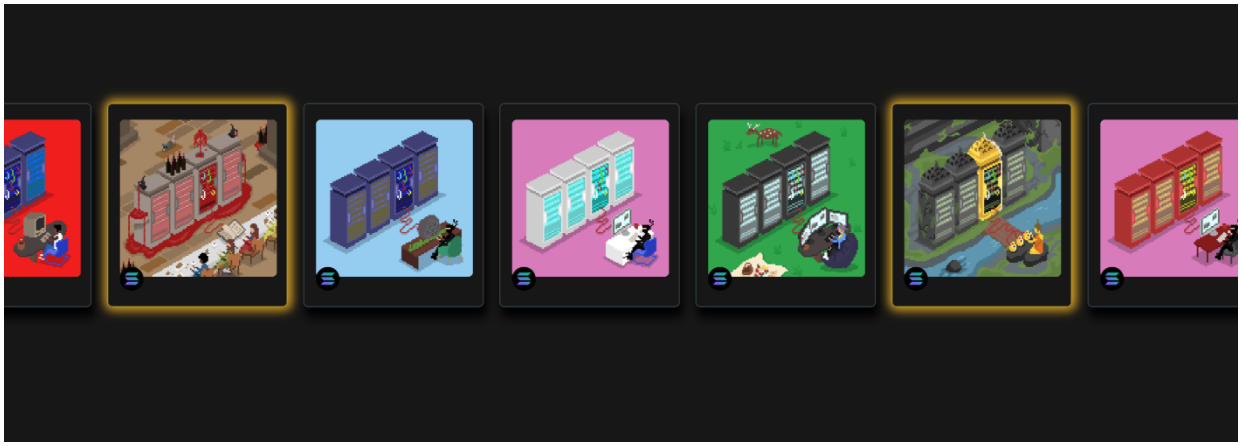
Additional Services

After the node successfully running, I still have plenty of resources on the Bare Metal server, so I decided to make this RPC project even more special by making 3 more webservices as one is already deployed as the solana-validator RPC response page, also multiple Windows RDP VMs.

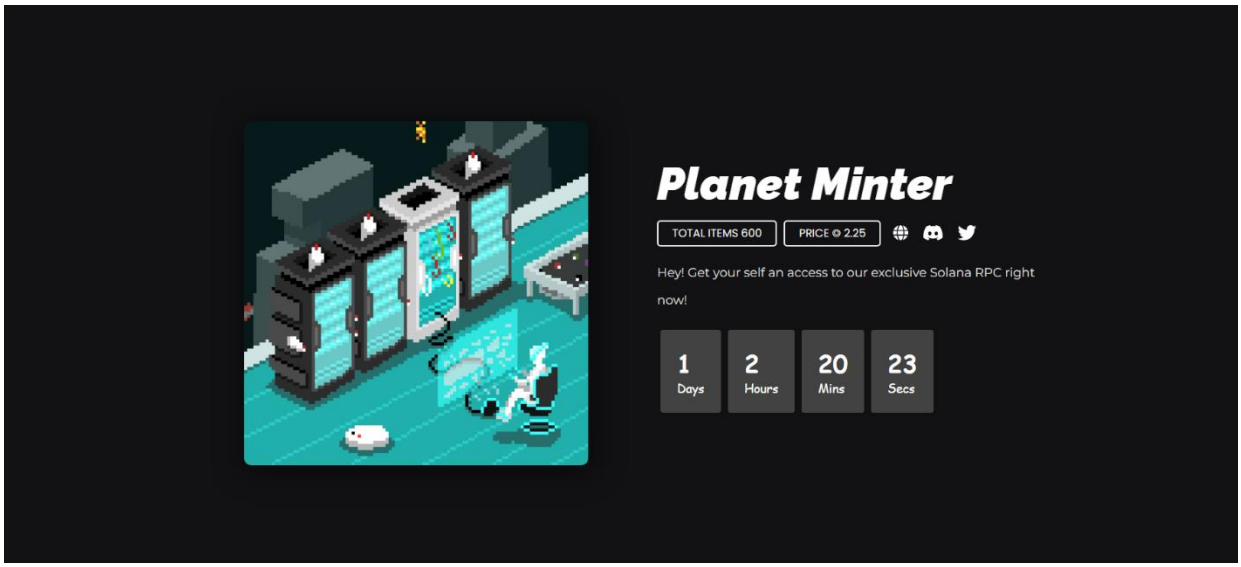
***Note**: Here is the solana-validator RPC response page by default, I bought planetminter.com domain and assign a subdomain to port 7000 of my node with NGINX reverse-proxy*

The three other webservices that I deployed was a Landing Page, a NFT holder verification page, and NFT minting page. Each assigned with subdomains of planetminter.com as shown below:

Home: goes to base url (home.planetminter.com)
Sneak Peek: goes to another page in home.planetminter.com
RPC Info: goes to another page in home.planetminter.com
About: goes to project info page in another page in home.planetminter.com
Verify Account: goes to verify.planetminter.com to verify NFT holder and automatic onboarding to discord bot
Mint: goes to mint.planetminter.com to buy or mint my deployed NFT to access the RPC node



***Note**: Sneak Peek page*



***Note**: mint.planetminter.com*

***Note**: verify.planetminter.com, I used and modified NFT4Cause verify tool, link below*

Ref: https://github.com/qrtp/solana-project-tools

Network and NGINX

So basically NGINX is the core thing in the project, with all the services inside a single server, NGINX was needed to do the routing from subdomains to the services in the local port and doing the whitelisting concept.

Things that I did was make all the services listens only in the local (not public), and ufw only allow necessary ports to be opened such as 8000-8020 (solana), and 443 (https), and multiple ports for Windows RDP VMs (will explain later).

From there, I create server block on the NGINX for each subdomain, with each server name assigned to subdomains I had to redirect with each local port for the specified services.

Example:

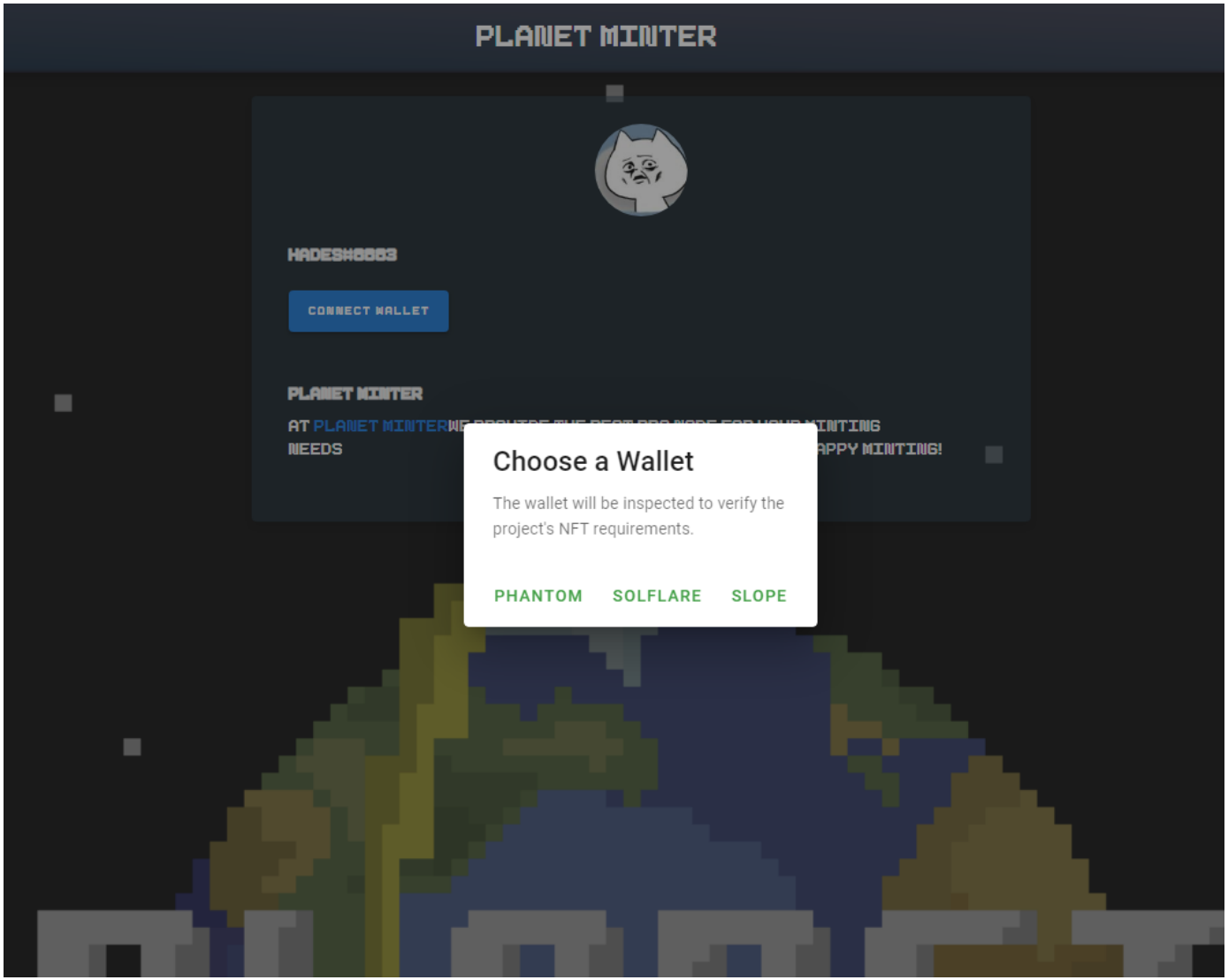
home.planetminter.com -> 127.0.0.1:3000
mint.planetminter.com -> 127.0.0.1:3001
verify.planetminter.com -> 127.0.0.1:3002

And the IP whitelisting was made easy with integration of Discord Bot and NGINX with help of UFW in linux, I basically denying almost all IP from my RPC port, and will allow someone's IP if they get whitelisted.

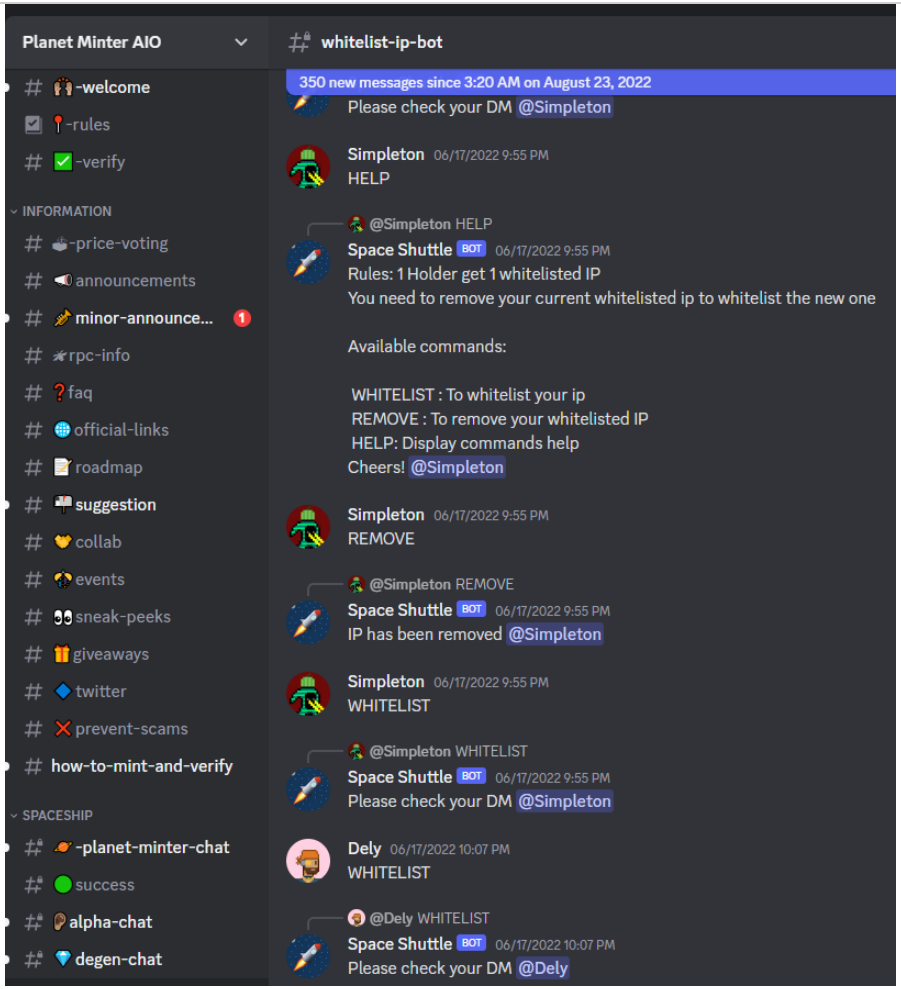
My previous verification page was made with integration of Solana, Phantom and Discord API. The process can be depicted below:

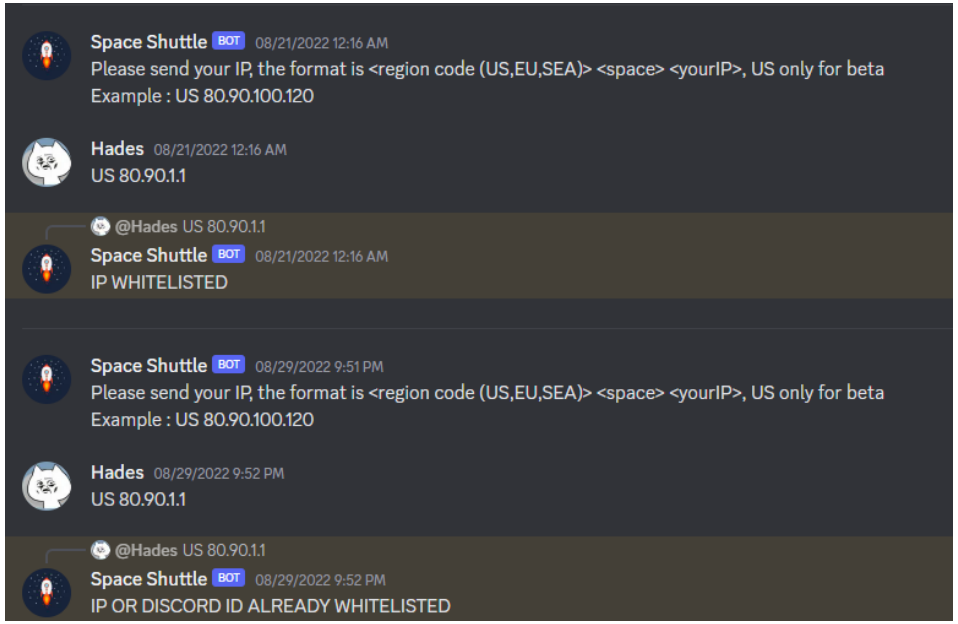
1. Bought the NFT at mint.planetminter.com
2. User connect their wallet and Discord account to verify at verify.planetminter.com
3. If user approved as NFT holder, discord role will be given to access the Whitelist bot in my Discord server.

The discord bot I made connect to my Bare-Metal Server and able to allow user's ip in the UFW and adding the ip to allowed ip database which NGINX will also allow the ip to redirect it to the RPC port in the local.

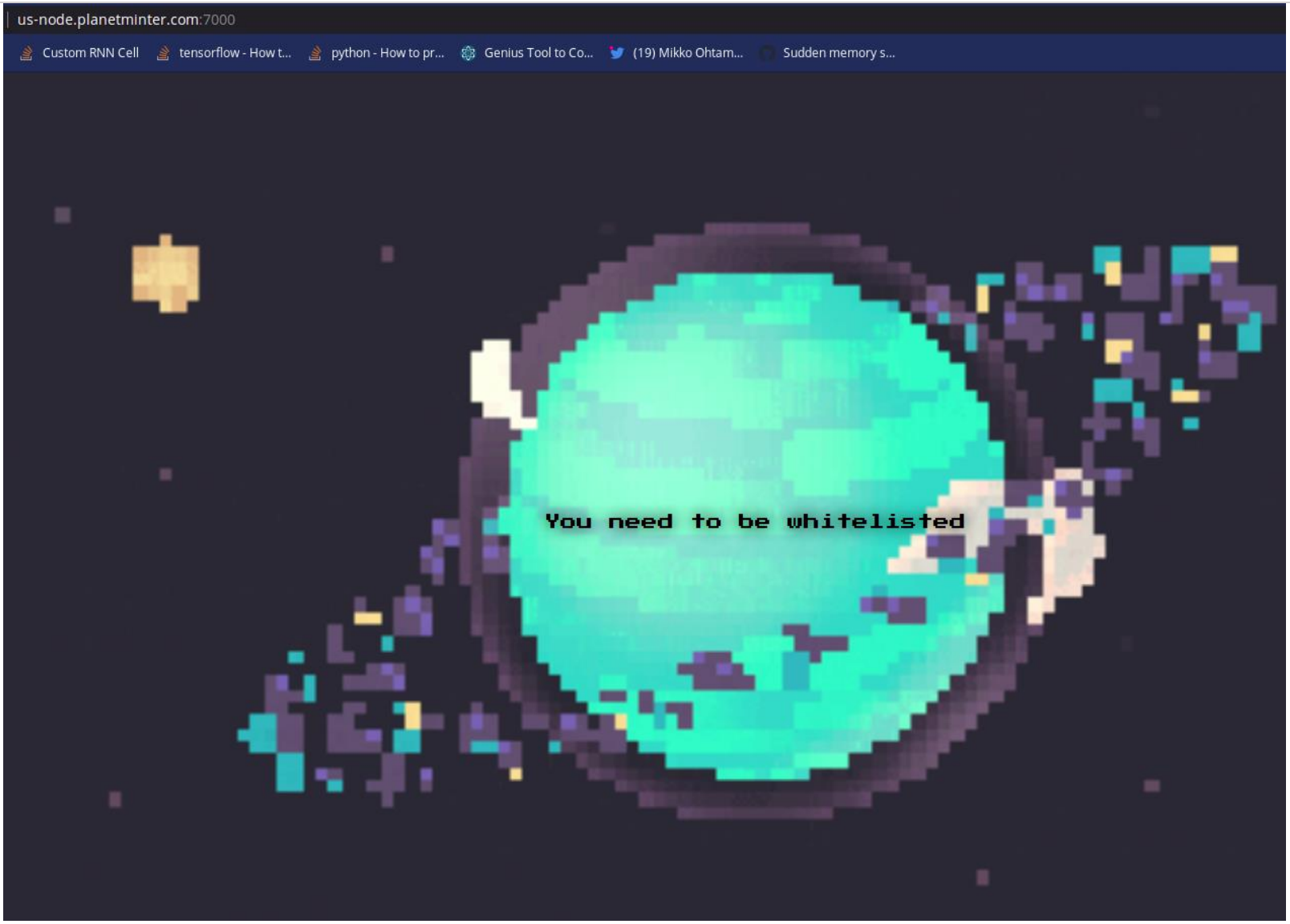


Note: *verify.planetminter.com, connect methods*





Note: *Discord Bot for IP whitelisting, coded in Python integrated with NGINX and ufw*



Note: *Gave a little touch on the blocked IP page, NGINX will redirect blocked IP to this page (it still use port 7000 on this image)*

Bonus services (Windows RDP VMs)

To gain more users on my RPC project, I offer 10 windows RDP servers which able to connect to my solana node locally for the first 10 NFT holders, so it will get the lowest latency (because it was on the same machine).

Because it was a Bare-Metal server, so I have virtualization capability on my machine, so I decided to use headless **virsh** and deployed multiple Windows 10 instances.

```
et-destroy       net-event         net-port-create   net-start
oot@planetminter:/mnt/vms/vm# virsh net-dhcp-leases default
Expiry Time           MAC address         Protocol   IP address           Hostname          Client ID or DUID
--------------------------------------------------------------------------------------------------------------------
2022-06-02 19:34:02   02:00:00:d1:78:d9   ipv4       192.168.122.41/24    DESKTOP-ROPB36A   01:02:00:00:d1:78:d9
2022-06-02 19:34:23   52:54:00:24:b0:18   ipv4       192.168.122.103/24   DESKTOP-ROPB36A   01:52:54:00:24:b0:18
2022-06-02 19:34:23   52:54:00:4b:1d:61   ipv4       192.168.122.123/24   DESKTOP-ROPB36A   01:52:54:00:4b:1d:61
2022-06-02 19:34:35   52:54:00:58:5b:1e   ipv4       192.168.122.19/24    DESKTOP-ROPB36A   01:52:54:00:58:5b:1e
2022-06-02 19:34:31   52:54:00:70:65:4a   ipv4       192.168.122.44/24    DESKTOP-ROPB36A   01:52:54:00:70:65:4a
2022-06-02 19:34:02   52:54:00:84:61:e9   ipv4       192.168.122.16/24    DESKTOP-ROPB36A   01:52:54:00:84:61:e9
2022-06-02 19:34:07   52:54:00:95:97:6f   ipv4       192.168.122.212/24   DESKTOP-ROPB36A   01:52:54:00:95:97:6f
2022-06-02 19:34:27   52:54:00:d7:0f:f4   ipv4       192.168.122.222/24   DESKTOP-ROPB36A   01:52:54:00:d7:0f:f4
2022-06-02 19:34:36   52:54:00:dd:bc:b8   ipv4       192.168.122.226/24   DESKTOP-ROPB36A   01:52:54:00:dd:bc:b8
2022-06-02 19:34:15   52:54:00:fd:21:e6   ipv4       192.168.122.102/24   DESKTOP-ROPB36A   01:52:54:00:fd:21:e6

oot@planetminter:/mnt/vms/vm# ^M
bash: $'\r': command not found
oot@planetminter:/mnt/vms/vm# iptables -t nat -I PREROUTING -p tcp -d 175.75.50.226 --dport 51 -j DNAT --to-destinatic
```

The instances uses NAT network on my bare-metal server as the host. As you can see DHCP is active and assigned their local IP automatically. Basically I made one instance, activate the RDP and open its 3389 port then clone it multiple times, as can be seen they all have the same hostname (can be renamed).

From there, I allocated 10 ports from my host network (public IP) for those VMs, each allocated port will redirect to each IP's 3389 port. The routing can be done with iptables.

Example:

iptables -t nat -I PREROUTING -p tcp -d 175.75.50.226 --dport 51 -j DNAT --to-destination 192.168.122.103:3389

Above command will route my public bare-metal IP (175.75.50.226) port 51 to one of the VM local ip (192.168.122.103) on port 3389, which is RDP service port for the VM.

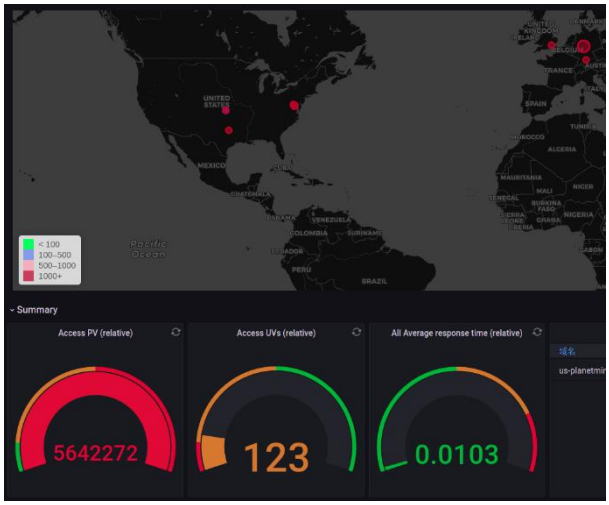
Server Monitoring (Prometheus, Grafana, NGINX)

Handling a blockchain node was never a simple thing when it comes to maintain its stability. In this case, Solana which keeps updating its system to a newer version like every few days or week, sometimes they urge the validator provider to update the node before specific date otherwise the node will be less functional, (plus I add a tons of other services underneath the node), things will sure be more challenging for me.

In order to made thing simpler, I installed prometheus and also make it available only on the local network. When prometheus ready, I need to get a better grip on how actually going on the server with some visualization, then I installed Grafana.

Grafana will receive prometheus data locally and put in the dashboard that I configured. The Grafana service I installed will be available on public network of the bare metal (of course with password), in case I was outside my desk and needed to check the node then I can just check it by accessing Grafana by any internet network I had even with smartphone browser.

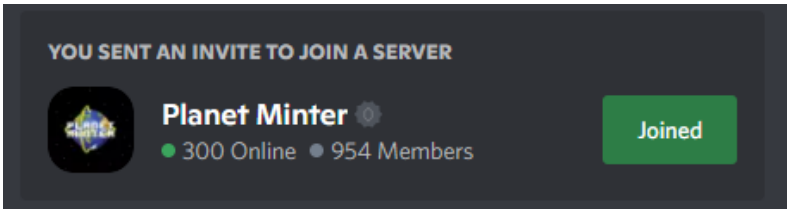
***Note**: Above is a screenshot on Grafana when the server about to crash from OOM*



Then I also add NGINX integration to gain visualization on users usage on my node. Each of the services including my solana-validator service will log their access and error log on separated log from each services. Promotheus also able to grab these logs and Grafana will visualize it right away. With this I can see how many users and their requests on my node.

Conclusion

Everything went well with the whole deployment. This was conducted mainly for educational purpose with a bit of business thing on top of it. The project gains almost two thousands of users (the screenshot of peak amount of members is missing).



I gain a ton of knowledge on many aspects. Here are the points I got from the project:

- High Performance Bare Metal managing (initiated with only SSH)
- Assessed my skill for fully headless server management
- Deployed Solana NFT
- Solana NFT holders verification without paid third party service
- Assessed my skills on NGINX, Grafana and Prometheus
- Assessed my skills on Network Management
- Assessed my skills on Web Development, Discord Bot, and Custom Scripting

success

which bot ? (edited)

clevermonster 06/25/2022 1:56 AM

5+ sol profit.... (Pingus Dao) ❤️ Used Planet Minter RPC

Primary | Social | Promotions 1 new
Groupon Goods

Magic Eden Team — New purchase for 0.24 SOL on Pingu #1480 - Someone purchas

success

EditEveryFrame 07/09/2022 2:02 AM

Neura + PlanetMinter Node, Transactions were immediate

Shrouded Playground 2

🔥 7 🧱 2

MrAnouar 08/27/2022 11:14 PM

very good success using Planetminter's node in high congested drops, thanks for the team

🔥 1

Edmund 08/29/2022 9:15 PM

Lucentals ITEMS 2 FLOOR 0.5 TOTAL 1.00

EditEveryFrame 06/15/2022 5:13 AM

Legit hit 5 in like 3 seconds, Neura + Plant Minter, such a good node

Deymar 06/03/2022 10:42 PM
hope y'all like it

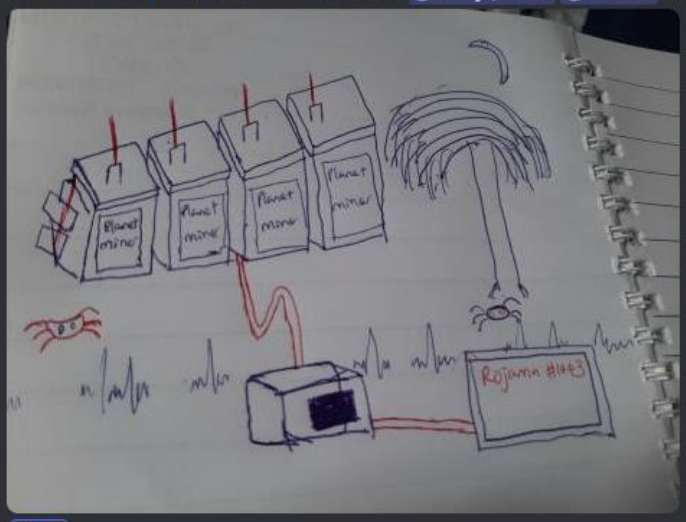
🅿lanet Minter is 🅿🥂

@Master

Deymar 06/03/2022 11:33 PM
@Milky | MOD

cryptXM 06/04/2022 12:49 AM
I'm not so good..hope my art on planet minter is good enough 😁 @Milky | MOD @Master 🚀

rojama 06/03/2022 9:01 PM
Not much of an artist but made a fan art @Milky | MOD @Master (edited)



😍 2

***Note**: The Project was discontinued*

*For verification you can go to **planetminter.com** for proof of me as project owner.*

Thanks to few of my business major friends who helped me with moderating the Discord Server and helped designing the NFT arts.

A lot of screenshots were missing, you can ask me further for the details.